

Mathematical Induction

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Learning Outcome

At the end of this lecture, you should be able to:

- Understand and apply the proof by mathematical induction.

Generalisation

Consider the following problem:

Can you write down a general rule for the sequence:

$1, 1 + 8, 1 + 8 + 27, 1 + 8 + 27 + 64, \dots$?

We can look at the above sequence as:

$$\begin{aligned}a_1 &= 1^3, \\a_2 &= 1^3 + 2^3 = a_1 + 2^3, \\a_3 &= 1^3 + 2^3 + 3^3 = a_2 + 3^3, \\a_4 &= 1^3 + 2^3 + 3^3 + 4^3 = a_3 + 4^3. \\&\vdots\end{aligned}$$

Each number in the sequence is obtained by adding the preceding term by adding the respective cube.

So, we can define this sequence **inductively** (or **recursively**) by $a_1 = 1$,
 $a_n = a_{n-1} + n^3$, for $n > 1$.

Generalisation

Example:

$$\begin{aligned} 1 &= 1^2, & \leftarrow n=1 \\ 1 + 3 &= 4 = 2^2, & \leftarrow n=2 \\ 1 + 3 + 5 &= 9 = 3^2, & \leftarrow n=3 \\ 1 + 3 + 5 + 7 &= 16 = 4^2, & \leftarrow n=4 \\ &\vdots & \leftarrow n=5 \\ &\vdots & \leftarrow n=6 \\ &\vdots & \vdots \end{aligned}$$

n^2 ↓

Find a generalisation for the above expressions.

$$\text{For } n=1, \quad 1^2 = 1 = (2(1)-1)$$

$$\text{For } n=2, \quad 2^2 = 4 = 1+3 = 1 + (2(2)-1)$$

$$\text{For } n=3, \quad 3^2 = 9 = 1+3+5 = 1+3 + (2(3)-1)$$

$$\text{For } n=4, \quad 4^2 = 16 = 1+3+5+7 = 1+3+5 + (2(4)-1)$$

Thus, we suspect that we could generalize this as
 $1+3+5+\dots+(2n-1) = n^2$, for $n \geq 1$

The Principle of Mathematical Induction

Mathematical induction is a method of proving a universal statement with a countable domain.

The purpose is to show that $P(n)$ is true for every positive integer n .

Basically, proving a statement by mathematical induction involving two steps

- ① Initial step (Base case)
- ② Inductive step (Assuming inductive hypothesis and showing it implies the subsequent case) $k^{\text{th}} \text{ case} \Rightarrow (k+1)^{\text{th}} \text{ case.}$

First Principle of Mathematical Induction

Let $P(n)$ be a predicate that is defined for positive integers n . Suppose that

- ① $P(1)$ is true.
- ② For all integers $k \geq 1$, if $P(k)$ is true, then $P(k + 1)$ is true.

Then $P(n)$ is true for all positive integers n .

$$P(1) \Rightarrow P(2) \Rightarrow P(3) \Rightarrow \dots$$

First Principle of Mathematical Induction

Example: Prove that for any $n \in \mathbb{Z}^+$, $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$

$$\sum_{i=1}^n i = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}.$$

Let $P(n) = \frac{\sum_{i=1}^n i = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}}$

① Initial Step: When $n=1$, $\sum_{i=1}^1 i = 1 = \frac{1(1+1)}{2}$

So, $P(1)$ is true.

② Inductive step: Assume that $P(k)$ is true for some $k \geq 1$

$\Rightarrow P(k) = \sum_{i=1}^k i = 1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$ is true

First Principle of Mathematical Induction

for some positive integer k .

(We want to show $P(k+1)$ is true)

When $n = k+1$, $\sum_{i=1}^{k+1} i = (1+2+\dots+k) + (k+1)$

$$= \frac{k(k+1)}{2} + k+1, \text{ from inductive hypothesis}$$

$$= \frac{k(k+1)}{2} + \frac{2(k+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

$$\text{So, } P(k+1) : \sum_{i=1}^{k+1} i = \frac{(k+1)(k+2)}{2} \text{ is true.}$$

Hence, by mathematical induction, $P(n)$ is true for all positive integer n .

First Principle of Mathematical Induction

Example: Prove that for any $n \in \mathbb{Z}^+, = \{1, 2, 3, \dots\}$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

$$\text{Let } P(n) : \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

① Initial step: When $n=1$, $P(1) : \sum_{i=1}^1 i^2 = 1^2 = \frac{1(1+1)(2(1)+1)}{6}$

So, $P(1)$ is true.

② Inductive step: Assume that $P(k)$ is true for some $k \geq 1$, i.e.

$$P(k) : \sum_{i=1}^k i^2 = 1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

is true for some positive integer k .

First Principle of Mathematical Induction

[we want to show that $P(k+1): \sum_{i=1}^{k+1} i^2 = \frac{(k+1)(k+2)(2(k+1)+1)}{6}$]

when $n=k+1$, $\sum_{i=1}^{k+1} i^2 = \underbrace{1+2^2+3^2+\dots+k^2}_{\substack{\text{is true. } (k+1)(k+2) \\ (2k+3)}} + (k+1)^2 \cdot \frac{(2k+3)}{6}$

$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$, from inductive hypothesis.

$= \frac{k(k+1)(2k+1)}{6} + \frac{6(k+1)^2}{6}$

$= \frac{(k+1)}{6} [k(2k+1) + 6(k+1)]$

$= \frac{(k+1)}{6} [2k^2 + k + 6k + 6]$

$= \frac{(k+1)}{6} (2k^2 + 7k + 6)$

$$= \frac{(k+1)}{6} [(k+3)(k+2)]$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

$$\text{So, } P(k+1) : \sum_{i=1}^{k+1} i^2 = \frac{(k+1)(k+2)(2k+3)}{6} \quad \text{B true.}$$

Hence, by mathematical induction, $P(n)$ is true for all positive integers n .

Extended Principle of Mathematical Induction

Let $P(n)$ be a predicate that is defined for integers $n \geq a$ where a is a fixed integer. Suppose that

- 1 $P(a)$ is true.
- 2 For all integers $k \geq a$, if $P(k)$ is true, then $P(k+1)$ is true.

Then $P(n)$ is true for all integers $n \geq a$.

$$p(a) \Rightarrow p(a+1) \Rightarrow p(a+2) \Rightarrow \dots$$

Extended Principle of Mathematical Induction

Example: Prove that $(n+1)! > 2^{n+3}$ for all integers $n \geq 5$. = {5, 6, 7, 8, ...}

Let $P(n) : (n+1)! > 2^{n+3}$

$6 \times 5 \times 4 \times 3 \times 2 \times 1$

① Initial step: When $n=5$, $(n+1)! = 6! = 720$
and $2^{5+3} = 256$.

So, $P(5) : (5+1)! > 2^{5+3}$ is true.

② Inductive step: Assume that $P(k)$ is true for some integer $k \geq 5$, i.e.

$P(k) : (k+1)! > 2^{k+3}$ is true for some integer $k \geq 5$

[We want to show that $P(k+1) : (k+2)! > 2^{k+4}$ is true]

$$\text{When } n = k+1, [(k+1)+1]! = (k+2) \underbrace{(k+1)k(k-1)\dots 3\cdot 2\cdot 1}_{(k+1)!}$$

$$> (k+2) 2^{k+3} \quad \text{from inductive hypothesis}$$

$$= k 2^{k+3} + 2 \cdot 2^{k+3}$$

$$= \underline{k 2^{k+3}} + 2^{k+4}$$

$$> 2^{k+4}$$

$$\text{So, } P(k+1) : [(k+1)+1]! > 2^{k+4} \text{ is true.}$$

Hence, by mathematical induction, $P(n)$ is true for all integers $n \geq 5$.

More Examples

Example: Show that $1^3 + 2^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$ for all integers $n \geq 1$.

Exercise.

More Examples

More Examples

Example: Show that $7^{2n} + 16n - 1$ is divisible by 64 for all integers $n \geq 0$.

Let $P(n): 7^{2n} + 16n - 1$ is divisible by 64.

$$[7^{2n} + 16n - 1 = 64m \text{ for some integer } m]$$

Initial step: When $n=0$, $7^{2(0)} + 16(0) - 1 = 0$ is divisible by 64. So $P(0)$ is true.

Inductive step: Assume that $P(k)$ is true for some integer $k \geq 0$
i.e. $7^{2k} + 16k - 1$ is divisible by 64 for some integer $k \geq 0$.
 $7^{2k} + 16k - 1 = 64m$ for some integer m .
 $\implies 7^{2k} = 64m - 16k + 1$

We want to show that $P(k+1): 7^{2(k+1)} + 16(k+1) - 1$ is divisible by 64.

More Examples

When $n = k+1$,

$$7^{2(k+1)} + 16(k+1) - 1 = 7^{2k+2} + 16k + 16 - 1$$

$$= 7^2 (7^{2k}) + 16k + 15$$

$$= 49(64m - 16k + 1) + 16k + 15$$

for some integer m ,
from inductive hypothesis

So, $7^{2(k+1)} + 16(k+1) - 1$
is divisible by 64 and
 $P(k+1)$ is true.

Hence, by the mathematical
induction, $P(n)$ is true

for all integers $n \geq 0$.

$$= 49(64m) - 49(16k) + 49 + 16k + 15$$

$$= 64(49m) - 48(16k) + 64$$

$$= 64[49m - 12k + 1], \text{ where } 49m - 12k + 1 \text{ is an integer.}$$

More Examples

Example: Show that $1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1} n^2 = (-1)^{n-1} \left(\frac{n(n+1)}{2} \right)$ for all integers $n \geq 1$.

$$\text{Let } p(n): 1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1} n^2 = \sum_{i=1}^n (-1)^{i-1} i^2$$
$$\longrightarrow = (-1)^{n-1} \left(\frac{n(n+1)}{2} \right)$$

$$\text{Initial step: When } n=1, \sum_{i=1}^1 \underline{(-1)^{i-1}} \underline{i^2} = 1 = (-1)^{1-1} \left(\frac{1(1+1)}{2} \right)$$

So, $P(1)$ is true.

Inductive step: Assume that $P(k)$ is true for some integer $k \geq 1$
i.e., $\sum_{i=1}^k (-1)^{i-1} i^2 = (-1)^{k-1} \left[\frac{k(k+1)}{2} \right]$ is true for
some integer $k \geq 1$.

More Examples

We wish to show that $P(k+1) = \sum_{i=1}^{k+1} (-1)^{i-1} i^2 = \underline{\underline{(-1)^k \left[\frac{(k+1)(k+2)}{2} \right]}}$ is true.

$$\text{When } n=k+1, \quad \sum_{i=1}^{k+1} (-1)^{i-1} i^2 = \sum_{i=1}^k (-1)^{i-1} i^2 + (-1)^{k+1-1} (k+1)^2$$

$$= (-1)^k \left[\frac{k(k+1)}{2} \right] + \frac{2(-1)^k (k+1)^2}{2} \quad \text{from inductive hypothesis}$$

$$\frac{(-1)^k}{-1} = \frac{(-1)^k k(k+1) + 2(-1)^k (k+1)^2}{2}$$

$$= \frac{(-1)^k (k+1) \left[\overset{k}{-k} + 2(k+1) \right]}{2}$$

$$= \frac{(-1)^k (k+1) (-k + 2k + 2)}{2}$$

$$= (-1)^k \left[\frac{(k+1) (k+2)}{2} \right]$$

So, $P(k+1)$ is true.

Hence, by the mathematical induction, $P(n)$ is true for all integers $n \geq 1$.

More Examples

Example: Show that $1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n! = (n+1)! - 1$ for all integers $n \geq 1$.

$$\text{Let } P(n) : \underline{1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n!} = \sum_{i=1}^n i \cdot i! \\ = (n+1)! - 1$$

Initial Step : $n=1, \sum_{i=1}^1 i \cdot i! = 1 \cdot 1! = 1 = (1+1)! - 1$
You have to actually check it.

So, (1) is true.

Inductive Step : Assume that $P(k)$ is true for some integer $k \geq 1$
i.e., $\sum_{i=1}^k i \cdot i! = (k+1)! - 1$ is true for some integer
 $k \geq 1$

More Examples

We wish to show that

$$p(k+1): \sum_{i=1}^{k+1} i \cdot i! = \underline{(k+2)! - 1} \text{ is true}$$

$$\text{When } n = k+1, \quad \sum_{i=1}^{k+1} i \cdot i! = \sum_{i=1}^k i \cdot i! + (k+1) \cdot (k+1)!$$

$$= (k+1)! - 1 + (k+1) \cdot (k+1)! \quad \text{from inductive hypothesis}$$

$$= \underline{(k+1)!} + \underline{(k+1) \cdot (k+1)!} - 1$$

$$= (k+1)! [1 + k+1] - 1$$

$$= \underline{(k+1)!} \cdot \underline{(k+2)} - 1$$

$$(k+2)(k+1)k(k-1)(k-2) \cdots 3 \cdot 2 \cdot 1 = (k+2)!$$

$$= (k+2)! - 1$$

So, $P(k+1)$ is true.

Hence, by the mathematical induction, $P(n)$ is true
for all integers $n \geq 1$.

More Examples

Example: Show that $1 + 3 + 5 + \dots + (2n - 1) = n^2$ for all integers $n \geq 1$.

$$\text{Let } P(n) = 1 + 3 + 5 + \dots + (2n - 1) = n^2$$

Initial Step: When $n=1$, $(2(1)-1) = 1 = 1^2$

So, $P(1)$ is true.

Inductive hypothesis: Assume that $P(k)$ is true for some integer $k \geq 1$
i.e., $1 + 3 + 5 + \dots + (2k - 1) = k^2$ is true for
some integer $k \geq 1$.

We wish to show that $P(k+1): 1 + 3 + 5 + \dots + (2k-1) + (2k+1) = \underline{\underline{(k+1)^2}}$
is true.

More Examples

When $n=k+1$, $\underline{1+3+5+\dots+(2k-1)+(2k+1)} = k^2 + (2k+1)$ from inductive hypothesis.

$$= (k+1)(k+1)$$
$$= (k+1)^2$$

So, $P(k+1)$ is true.

Hence, by the mathematical induction, $P(n)$ is true for all integers $n \geq 1$.